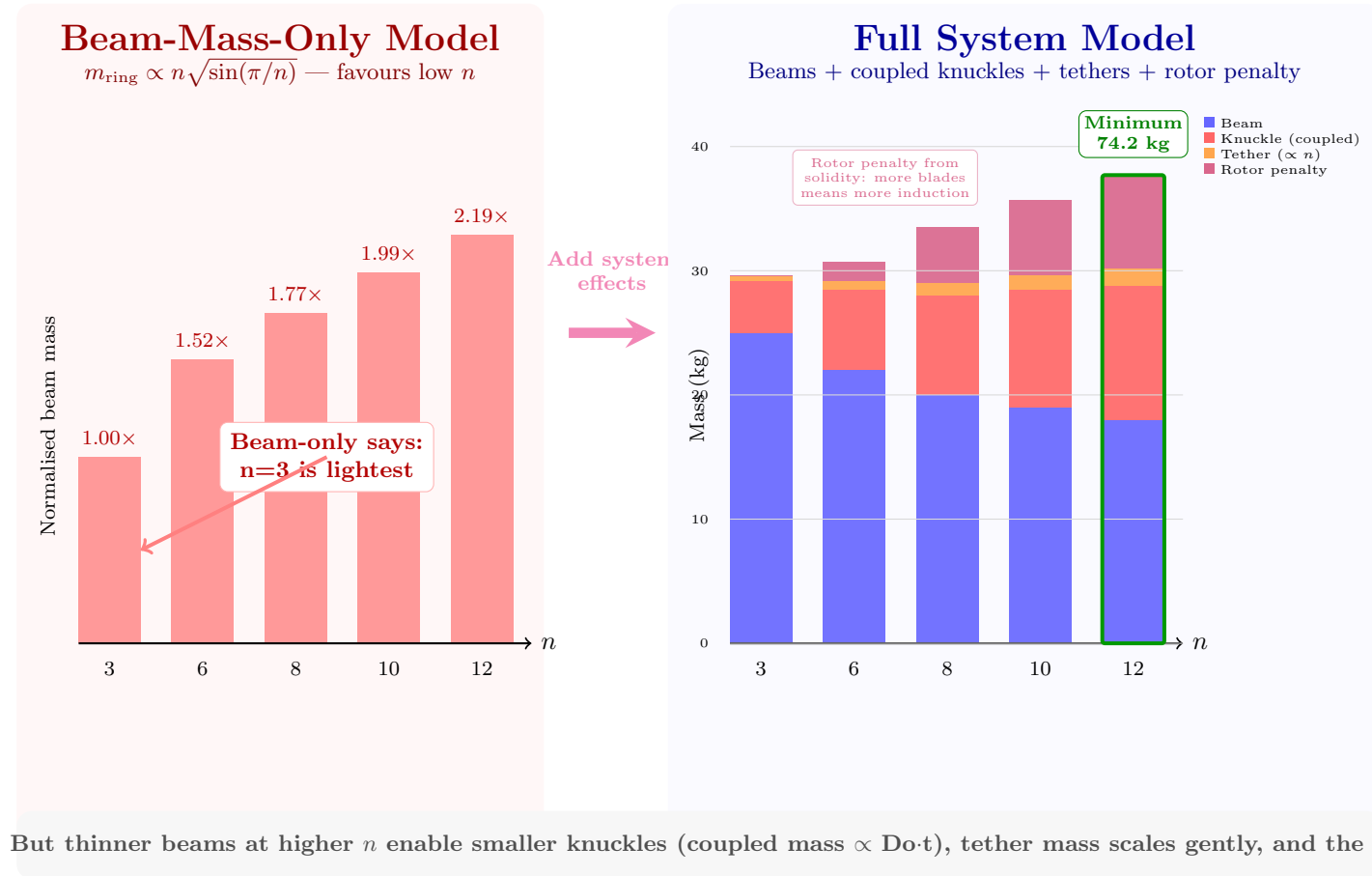


Why the Optimiser Chose $n = 12$ Despite Beam-Only Scaling

Beam mass alone says fewer lines are lighter. System effects reverse the tradeoff.



The counterintuitive result: Beam-only scaling says $n = 3$ is lightest. But thinner beams at higher n enable smaller knuckles (coupled mass $\propto D \cdot t$), tether mass scales gently, and the rotor solidity penalty is modest. The system optimum lands at $n = 12$.